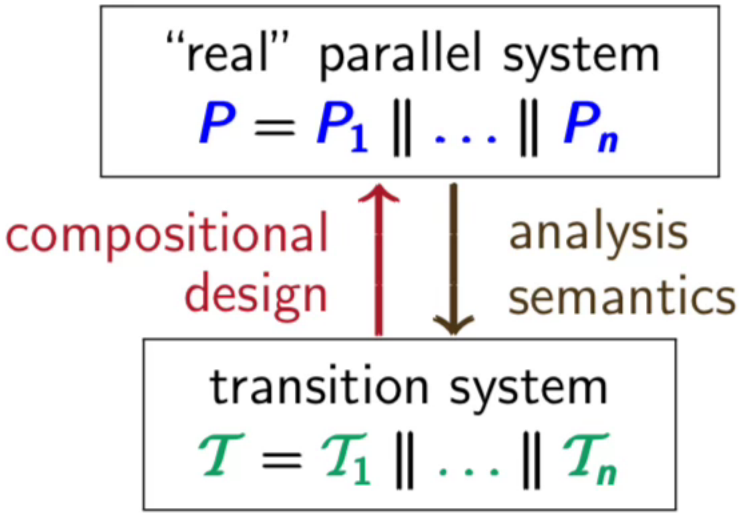
17/9/25



goal: define semantic parallel operators on transition systems or program graphs that model "real" parallel operators

$$\text{effect}(\alpha \underbrace{|||}_{\text{interleaving operator}} \beta) = \text{effect}(\alpha;\beta + \beta;\alpha)$$

parallel execution of α and β on *two processors* $\widehat{=}$ serial execution on a *single processor* in arbitrary order

$$\mathcal{T}_1 = (S_1, Act_1, \longrightarrow_1, S_{0,1}, AP_1, L_1)$$

$$\mathcal{T}_2 = (S_2, Act_2, \longrightarrow_2, S_{0,2}, AP_2, L_2)$$

$L : S \to 2^{AP}$

The transition system $\mathcal{T}_1 \,|||\, \mathcal{T}_2$ is defined by:

$$\mathcal{T}_1 \,|||\, \mathcal{T}_2 = (S_1 \times S_2, Act_1 \cup Act_2, \longrightarrow, S_{0,1} \times S_{0,2}, AP, L)$$

where the transition relation $\longrightarrow$ is given by:

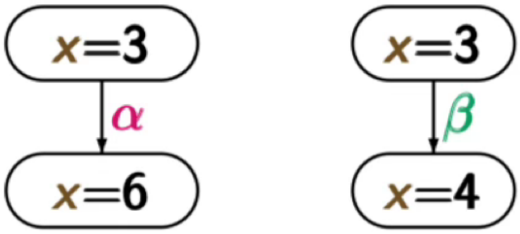
$$\frac{s_1 \xrightarrow{\alpha}_1 s_1'}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s_1', s_2 \rangle} \qquad \frac{s_2 \xrightarrow{\alpha}_2 s_2'}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s_1, s_2' \rangle}$$

atomic propositions: $AP = AP_1 \uplus AP_2$

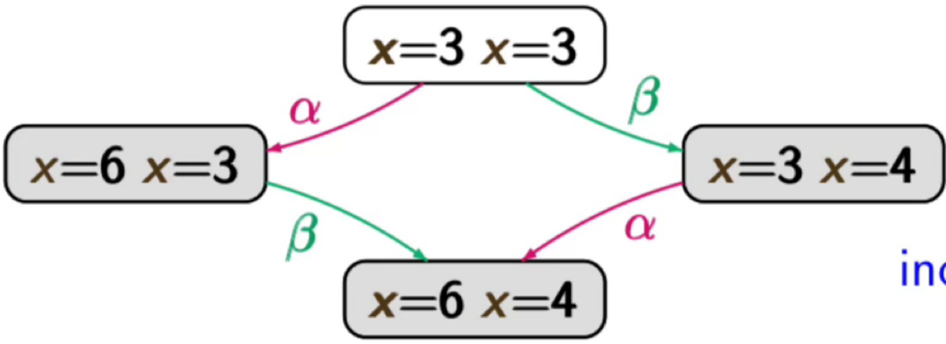
labeling function: $L(\langle s_1, s_2 \rangle) = L_1(s_1) \cup L_2(s_2)$

dependent actions $\alpha \mathrel{\widehat{=}} x{:=}2x$ and $\beta \mathrel{\widehat{=}} x{:=}x{+}1$

representations in transition systems



interleaving operator ||| for transition systems "fails"



inconsistent states

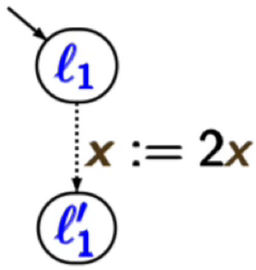
program graph $\mathcal{P}_1$ $(Loc_1, \ldots, \hookrightarrow_1, \ldots)$	program graph $\mathcal{P}_2$ $(Loc_2, \ldots, \hookrightarrow_2, \ldots)$

interleaving operator

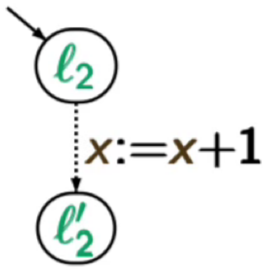
$$\mathcal{P}_1 \,|||\, \mathcal{P}_2 \;=\; (Loc_1 \times Loc_2, \ldots, \hookrightarrow, \ldots)$$

$$\frac{\ell_1 \overset{g:\alpha}{\hookrightarrow}_1 \ell_1'}{\langle \ell_1, \ell_2 \rangle \overset{g:\alpha}{\hookrightarrow} \langle \ell_1', \ell_2 \rangle} \qquad \frac{\ell_2 \overset{g:\alpha}{\hookrightarrow}_2 \ell_2'}{\langle \ell_1, \ell_2 \rangle \overset{g:\alpha}{\hookrightarrow} \langle \ell_1, \ell_2' \rangle}$$

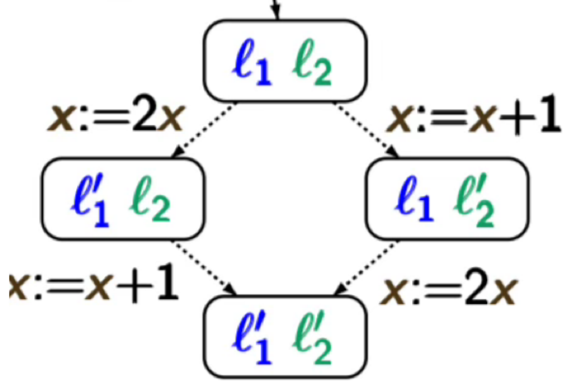
$\mathcal{P}_1$



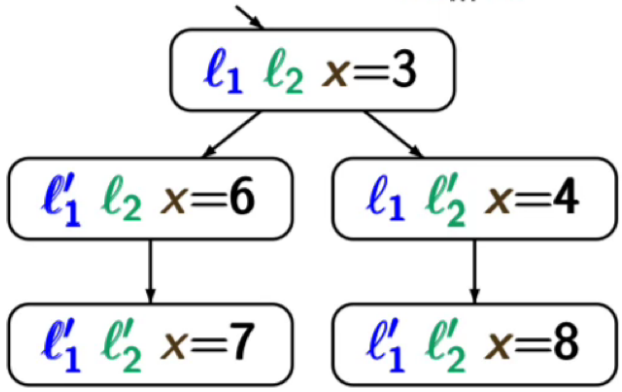
$\mathcal{P}_2$



$\mathcal{P}_1 \,|||\, \mathcal{P}_2$



transition system $T_{\mathcal{P}_1 ||| \mathcal{P}_2}$



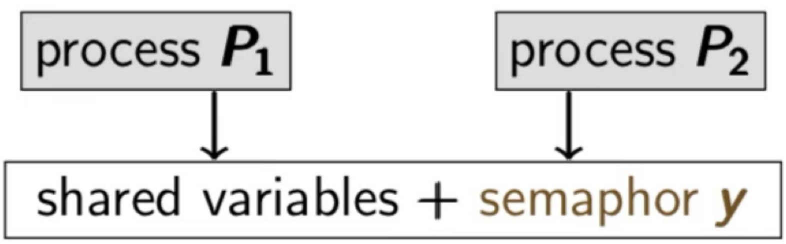
$$T_{P_1 ||| P_2} \neq T_{P_1} \,|||\, T_{P_2}$$

$$PG_i = (Loc_i, Act_i, Effect_i, \hookrightarrow_i, Loc_{0,i}, g_{0,i})$$

$$PG_1 \,|||\, PG_2 \;=\; (Loc_1 \times Loc_2, Act_1 \uplus Act_2, Effect, \hookrightarrow, Loc_{0,1} \times Loc_{0,2}, g_{0,1} \wedge g_{0,2})$$

$$\frac{\ell_1 \overset{g:\alpha}{\hookrightarrow}_1 \ell_1'}{\langle \ell_1, \ell_2 \rangle \overset{g:\alpha}{\hookrightarrow} \langle \ell_1', \ell_2 \rangle} \quad \text{and} \quad \frac{\ell_2 \overset{g:\alpha}{\hookrightarrow}_2 \ell_2'}{\langle \ell_1, \ell_2 \rangle \overset{g:\alpha}{\hookrightarrow} \langle \ell_1, \ell_2' \rangle}$$

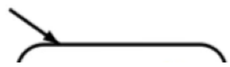
$$Effect(\alpha, \eta)(v) = \begin{cases} Effect_i(\alpha, \eta|_{Var_i})(v) & \text{if } v \in Var_i \\ \eta(v) & \text{otherwise} \end{cases}$$



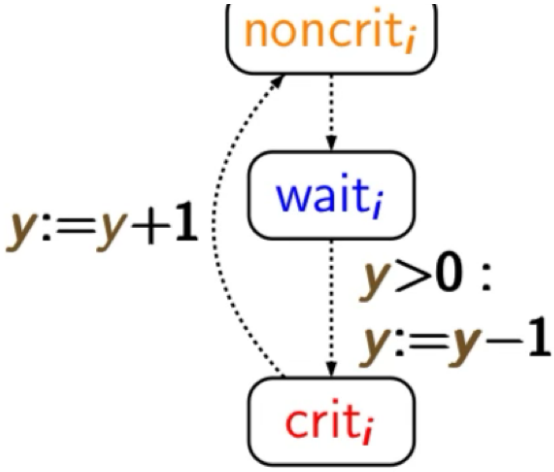
protocol for process P_i

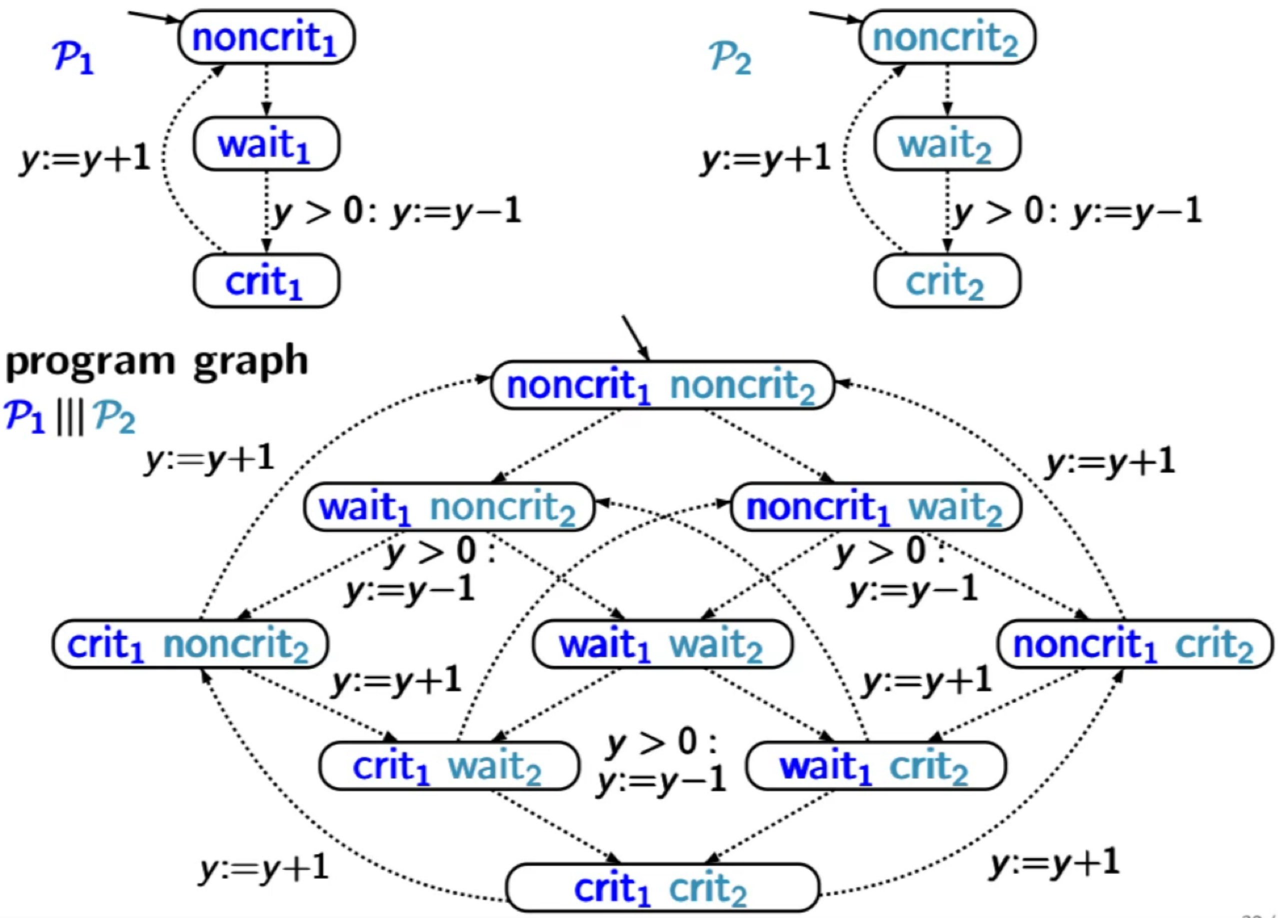
```
LOOP FOREVER
```

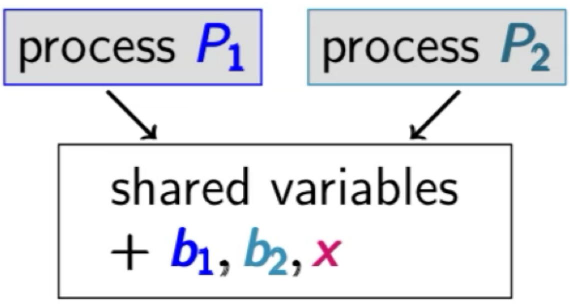
program graph $\mathcal{P}_i$

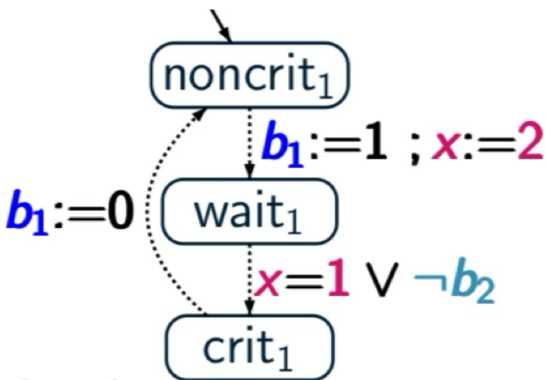


```
  noncritical actions;
  AWAIT  y > 0 DO
         y:=y−1
  OD
  critical actions;
  y:=y+1
END LOOP
```









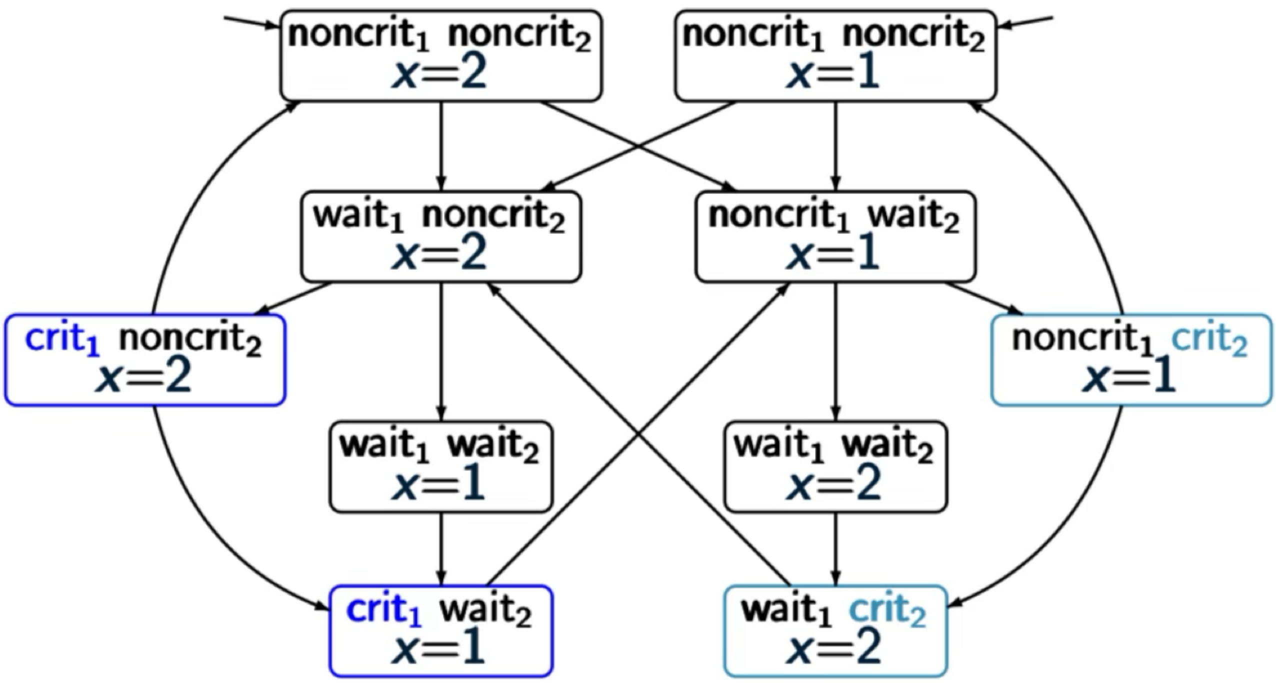
b_1, b_2 Boolean variables, $x \in \{1, 2\}$

```
LOOP FOREVER                              (* protocol for P1 *)
```

```
    noncritical actions;
    atomic{b1:=1 ; x:=2};
    AWAIT x=1 ∨ ¬b2 DO critical section OD
    b1:=0
END LOOP
```

TS for the Peterson algorithm

PC2.2-14



value of b_1 is given by **wait$_1$** $\vee$ **crit$_1$**
value of b_2 is given by **wait$_2$** $\vee$ **crit$_2$**

+ unreachable states

- true concurrency: interleaving operator $|||$ for TS (no communication, no dependencies)
- communication via shared variables
  - description of subsystems by program graphs
  - interleaving $|||$ for program graphs
  - TS is obtained by "unfolding"
- synchronous message passing $\longleftarrow$ data abstract
  - operator $\|_{Syn}$ for TS

* interleaving for independent actions
* synchronization over actions in *Syn*

- channel systems
  communication via shared variables + via channels

Synchronous message passing

PC2.2-17

$\mathcal{T}_1 = (S_1, Act_1, \rightarrow_1, \ldots)$, $\mathcal{T}_2 = (S_2, Act_2, \rightarrow_2, \ldots)$ TS

$Syn \subseteq Act_1 \cap Act_2$ set of synchronization actions

composite transition system:

$$\mathcal{T}_1 \|_{Syn} \mathcal{T}_2 = (S_1 \times S_2, Act_1 \cup Act_2, \rightarrow, \ldots)$$

interleaving for all actions $\alpha \in Act_i \setminus Syn$:

$$\frac{s_1 \xrightarrow{\alpha}_1 s_1'}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s_1', s_2 \rangle} \qquad \frac{s_2 \xrightarrow{\alpha}_2 s_2'}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s_1, s_2' \rangle}$$

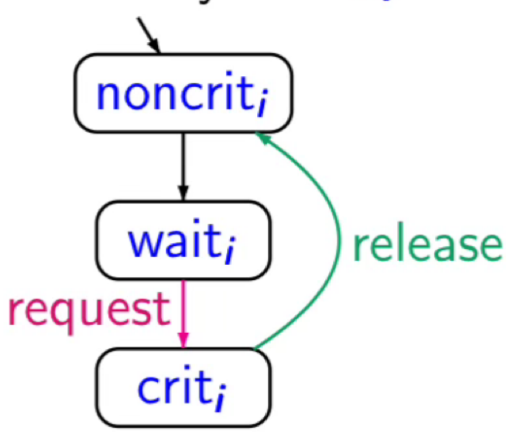
handshaking (rendezvous) for all $\alpha \in Syn$:

$$\frac{s_1 \xrightarrow{\alpha}_1 s_1' \quad \wedge \quad s_2 \xrightarrow{\alpha}_2 s_2'}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s_1', s_2' \rangle}$$

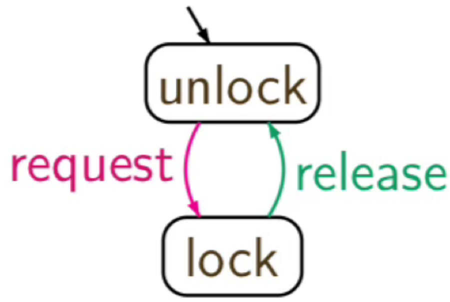
protocol for process P_i

```
LOOP FOREVER DO
      noncritical actions
      request
      critical section
      release
      noncritical actions
OD
```

transition system $\mathcal{T}_i$



Arbiter:
selects nondeterministically
a synchronization partner
$\mathcal{T}_1$ or $\mathcal{T}_2$



$$
\left.
\begin{array}{lcl}
\mathcal{T}_1 & = & (S_1, Act_1, \rightarrow_1, \ldots) \\
\mathcal{T}_2 & = & (S_2, Act_2, \rightarrow_2, \ldots) \\
\mathcal{T}_3 & = & (S_3, Act_3, \rightarrow_3, \ldots) \\
\mathcal{T}_4 & = & (S_4, Act_4, \rightarrow_4, \ldots) \\
\vdots & & \quad\quad \vdots
\end{array}
\right\} \text{transition systems}
$$

for $Syn \subseteq Act_1 \cup Act_2 \cup Act_3 \cup Act_4 \cup \ldots$

$$
\mathcal{T}_1 \|_{Syn} \mathcal{T}_2 \|_{Syn} \mathcal{T}_3 \|_{Syn} \mathcal{T}_4 \|_{Syn} \ldots \stackrel{\text{def}}{=}
\left(((\mathcal{T}_1 \|_{Syn} \mathcal{T}_2) \|_{Syn} \mathcal{T}_3) \|_{Syn} \mathcal{T}_4\right) \|_{Syn} \ldots
$$

where, e.g., $\mathcal{T}_1 \|_{Syn} \mathcal{T}_2 \stackrel{\text{def}}{=} \mathcal{T}_1 \|_{H} \mathcal{T}_2$

with $H = Syn \cap Act_1 \cap Act_2$

Parallel operator $\|$

PC2.2-OP-PAR

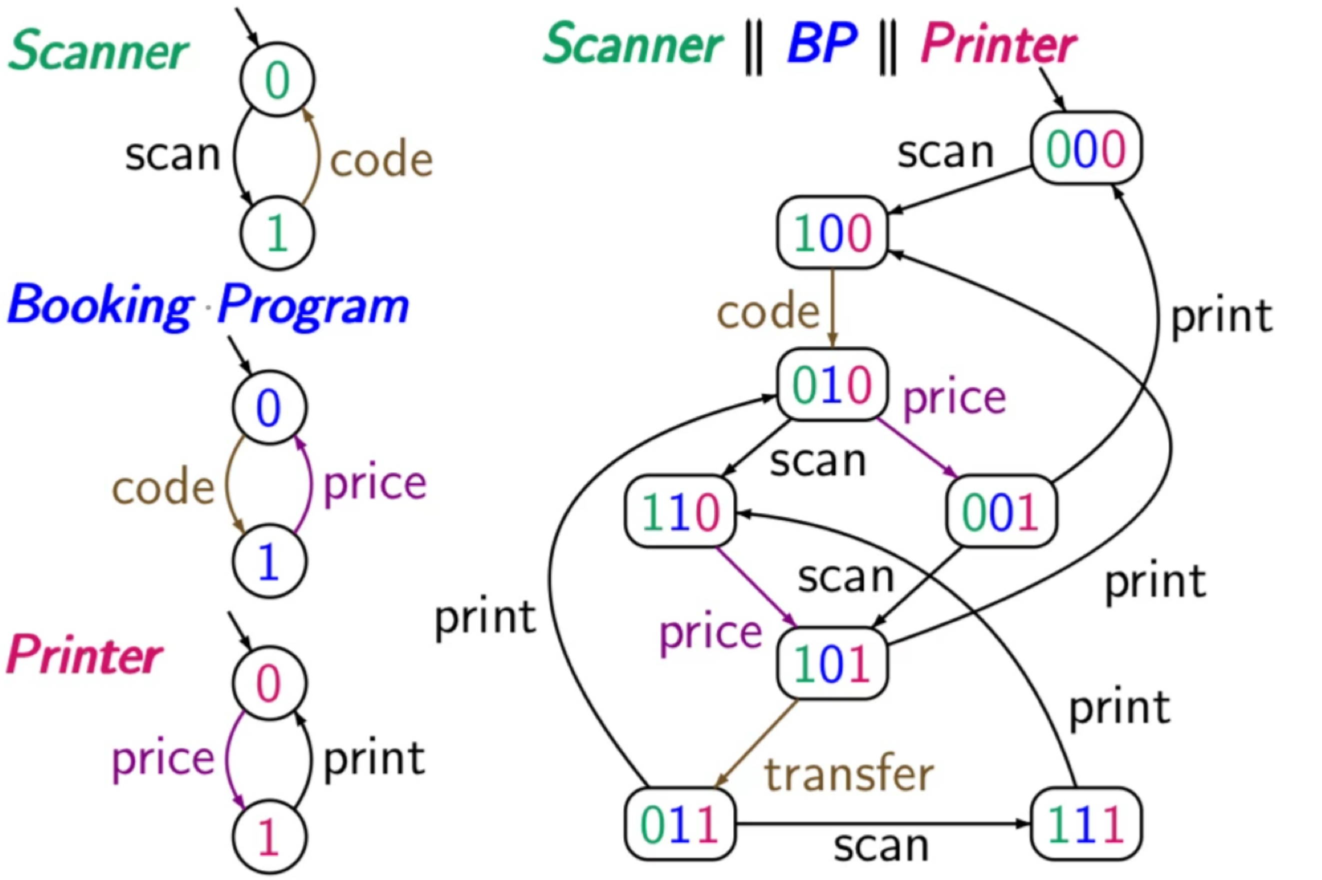
$$
\begin{array}{lcl}
\mathcal{T}_1 & = & (S_1, Act_1, \rightarrow_1, \ldots) \\
\mathcal{T}_2 & = & (S_2, Act_2, \rightarrow_2, \ldots) \\
\mathcal{T}_3 & = & (S_3, Act_3, \rightarrow_3, \ldots) \\
\mathcal{T}_4 & = & (S_4, Act_4, \rightarrow_4, \ldots) \\
\vdots & & \quad\quad \vdots
\end{array}
$$

transition systems s.t.
$Act_i \cap Act_j \cap Act_k = \emptyset$
if i, j, k are pairwise distinct

$$
\mathcal{T}_1 \| \mathcal{T}_2 \| \mathcal{T}_3 \| \mathcal{T}_4 \| \ldots \stackrel{\text{def}}{=}
$$

$$(((T_1 \,\|_{Syn_{1,2}}\, T_2) \,\|_{Syn_{1,2,3}}\, T_3) \,\|_{Syn_{1,2,3,4}}\, T_4) \cdots$$

$$\text{where}\quad \begin{aligned} Syn_{1,2} &= Act_1 \cap Act_2 \\ Syn_{1,2,3} &= (Act_1 \cup Act_2) \cap Act_3 \\ Syn_{1,2,3,4} &= (Act_1 \cup Act_2 \cup Act_3) \cap Act_4 \\ &\ \ \vdots \qquad\qquad \vdots \end{aligned}$$



$$(T_1 \,\|_{syn_1}\, T_2) \,\|_{syn_2}\, T_3$$

$$\not\simeq$$

$$T_1 \,\|_{syn_1}\, (T_2 \,\|_{syn_2}\, T_3)$$